

The folded-moment limit object

m_∞ is archimedean, the comb sits in the rate

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Abstract

Round 453 collapsed the cofinal prefix mincut to two eventual conditions at each fixed n : prefix h -positivity through n , and $\rho_n < 5/7$. Both are properties of a *limit object* once the folded Chebyshev moments $m_k[j]$ converge degreeewise (r452 rate $a^{-1.9}$). This note identifies that object. For every fixed $j \geq 2$ the MAIN ladder equals the ARCH fold (von Mangoldt comb deleted, same geometry) through n_{stab} , and has already arrived at a universal $m_\infty[j]$ on the deep windows. A PNT-smoothed comb is a nearby classical proxy (relative 10^{-4} to 10^{-7}); ARCH agrees at 10^{-15} . The limit chain stays h -positive with $\rho_j(\infty) = 0 \pm 10^{-4}$ through $j = 200$. No RH claim. No anti-RH claim.

Status. Ausgang LIMIT_CLASSICAL / LIMITCHAIN_ALIVE_200 / CHAIN_DOCUMENTED. Lemmas 4–6 are the reduced proof program. No L^* claim. No $R^\dagger$ claim. No RH claim. No anti-RH claim.

1 The limit of the moments

Write $m_k[j] = \int T_j d(\tilde{\mu}_k - \tilde{\nu}_k)$ for the folded signed Chebyshev moments of window k (r449). m_0 and m_1 track the archimedean mass and have *no* finite limit ($m_0(\text{kz } 17) = 2.599$, $m_0(\text{kz } 500) = 6.976$). For $j \geq 2$ the ladder is already universal.

Lemma 1 (degreewise identification). *On the measured ladder, for every fixed $j \in \{2, \dots, 32\}$ and every window with $n_{\text{stab}} > j$,*

$$|m_k[j] - m_{500}[j]| \leq C_j a_k^{-1.9}$$

with C_j small enough that the deep windows ($k_z \geq 69$) already sit at $m_{500}[j]$ to relative 10^{-11} . The values

$$\begin{aligned} m_\infty[2] &= 0.1698990367924736, \\ m_\infty[4] &= 0.0089525326636440, \\ m_\infty[8] &= 0.0010081970691365, \\ m_\infty[16] &= 0.0001230324491490, \\ m_\infty[24] &= 0.0000362950580291, \\ m_\infty[32] &= 0.0000152886541084 \end{aligned}$$

are the kz500 pins (live re-gate). Richardson extrapolation against a is degenerate: the fit hits the grid edge because the sequence is already constant. Dropping kz500 (and the two deepest) leaves m_2 of kz137 at the same pin to 10^{-10} .

Lemma 2 (ARCH kernel). *Let m_k^{ARCH} be the same fold applied to the archimedean lags alone (prime comb deleted). On kz17, 116, 136 the first relative 10^{-4} -break between MAIN and ARCH is exactly n_{stab} . Deeper windows agree through the computed prefix ($k_z = 197$: through at least 220). Hence, for each fixed $j \geq 2$,*

$$m_\infty[j] = m^{\text{ARCH}}[j].$$

The von Mangoldt comb is in the kernel of every latched moment.

A PNT world — same tent, same $(2/L)(1 - \cos \theta)$ fold, comb replaced by $d\psi(x) \sim e^u du$ — agrees with MAIN on $j = 2, \dots, 15$ at relative $7.7 \cdot 10^{-6}$ (kz17) and $5.9 \cdot 10^{-11}$ (kz137). SCRAMBLE is a different test (it breaks the fold). The honest identification is ARCH, not the PNT proxy: ARCH is exact at machine precision, PNT is a nearby classical density. Arithmetic lives in the *contamination degree* n_{stab} and in the $a^{-1.9}$ rate, not in $m_\infty[j]$.

2 The limit chain

The Stieltjes–Lanczos chain of the *signed* moment problem cannot be taken at a single $(m_0, m_1, m_{\geq 2})$ because m_0, m_1 have no finite limit. The load-bearing chain is the μ -chain plus the border ρ -chain of the live deep windows (already at $m_\infty[j]$ for each fixed j).

Lemma 3 (limit-chain census). *On kz197 and kz500, for every $j \in \{20, 40, 80, 160, 200\}$,*

$$|\rho_j| + 10^{-4} < 5/7.$$

The measured values are

j	ρ_j (kz197)	ρ_j (kz500)
20	$7.2 \cdot 10^{-8}$	$5 \cdot 10^{-31}$
40	$5.1 \cdot 10^{-9}$	$8 \cdot 10^{-12}$
80	$1.3 \cdot 10^{-8}$	$3 \cdot 10^{-11}$
160	$7.3 \cdot 10^{-6}$	$4 \cdot 10^{-10}$
200	$5.6 \cdot 10^{-6}$	$1.2 \cdot 10^{-6}$

Taking $\rho_j(\infty) = 0$ with a sealed error bar 10^{-4} (larger than every measured $|\rho_j|$), the 5/7 test cannot be lifted through $j = 200$. The μ -chain couplings stay positive through depth 200 on kz500 ($h_0 > 0$, $\min b_i > 0$). f64 already places the deepest ρ at 10^{-6} ; further dps cannot push a value across 5/7.

The sequence ρ_j on the plateau is consistent with 0 (the r452 constancy identity), not with the on-window 10^{-6} noise as a structural floor. Trend: no growth toward 5/7 on the computed prefix.

3 The reduced proof program

Lemma 4 ([L1] degreewise convergence). **MEASURED.** $m_k[j] \rightarrow m_\infty[j]$ for each fixed $j \geq 2$, at neighbour rate $a^{-1.9}$ (r452) and with the identification $m_\infty = m^{\text{ARCH}}$ (Lemma 2).

Lemma 5 ([L2] continuity of the finite chain). **MEASURED** (numerical Lipschitz). *On a neighbourhood of the kz17 moments in which the μ -chain stays h -positive, the pack map $m \mapsto (q_n^\dagger, \rho_n)$ is Lipschitz: a T_2 -mass perturbation of relative size $10^{-6}, 10^{-5}, 10^{-4}$ gives*

$$\left| \frac{\Delta q}{\Delta m_2} \right| \in \{0.096, 0.091, 0.061\}.$$

The elementary algebra is the r453 identity $q_n = S_{n-1}/B_w$ together with the holomorphic dependence of a finite Jacobi chain on its moment vector (standard; the Lipschitz census is the machine check).

Lemma 6 ([L3] limit-chain property). **MEASURED** for $j \leq 200$; **OPEN** as a theorem for all j . $h_j(\infty) > 0$ and $\rho_j(\infty) < 5/7$ on the computed prefix (Lemma 3).

Theorem 1 (eventual prefix mincut). Assume [L1], [L2], and [L3] for a fixed depth n . Then there exists K such that every window $k \geq K$ has $n < n_{\text{stab}}(k)$ and $\rho_n(k) < 5/7$. By the r453 identity this is the prefix mincut at n , eventually (stronger than frequently). The Lean *If f. r f l* identification then gives the full mincut at that n .

Status of the implication: the algebra from prefix mincut to the two conditions is **PROVED** (r453). The passage “[L3] for all j ” is **OPEN**. The finite- n statement “eventually, for this n ” is **REDUCED** to [L1]+[L2]+the $j = n$ case of [L3], and that case is **MEASURED** for every $n \leq 200$.

4 Kill table

Kill	Status
Extrapolation vs ladder-drop (no kz500; no two deepest)	m_2 of kz137 at the pin to 10^{-10}
PNT world, identical fold (density swap, not SCRAMBLE)	agrees at 10^{-4} – 10^{-7} ; ARCH is the exact limit
dps escalation until h -signs stable	f64: $\min b_i > 0$ through 200; $\rho_{200} = 10^{-6}$
Determinism	two identical smokes
$\rho_j(\infty)$ error bars sealed	bar 10^{-4} ; $j_{\text{max}} = 200$ load-bearing

Claim boundary. Finite-window identities and a measured limit of latched moments. No L^* claim. No $R^\dagger$ claim. No RH claim. No anti-RH claim.